Lec 1: Project management

- **Project planning and scheduling.**

Project managers are responsible for planning, estimating and scheduling project development and assigning people to tasks

modereen el masharee3 mas2oleen 3an ta5teet .. w ta2deer .. w gadwalet .. tatweer el mashroo3 .. w esnad el mohemmat lel naas
y3ni lamma ykono 3ayzeen yetawaro mashhroo3 .. l modereen homma el bey5atato wey2adaro weygadwelo el tatweer dah .. w kman bey2asemo el mahaam 3la el nas .. enta ta5od el goz2 dah wenta ta5od el goz2 dah wahakaza

- **Critical path: longest path in time. (short path)**

□ **Path that will cause entire project to fall behind if one day's delay is encountered.**

el critical path hwa el path aw el masar el bey2addi bel mashroo3 kollo l2enno yes2at law el mashro3 da wageh ta25er youm wa7ed bas

□ **The critical path is the sequence of dependent activities that defines the time required to complete the project.**

□ **It determines the project duration.**

el critical path da 3bara 3an selsela mn el ansheta el mo3tameda 3la ba3daha .. wbeha ba3raf el wa2t elle me7tageeno 3shan el project ye5las

bet7aded moddet el mashro3 (el modda el haye5las feha el mashro3)

□ **Any slippage in the completion of any critical activity causes project delays.**

slippage = enzelaq / tafweet

ay enzelaq aw tafweet ye7sal f 3amaleyet el completion bta3et el critical activity .. beysabeb ta25eer lel project

wel kalam da 3'aleban maygeesh 3aleh so2al nazari .. hwa so2alo mas2ala

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- **What are the attributes of good software?**

Eh el 7gat el t3arrafak en el software dah GOOD

• The software should deliver the required functionality and performance to the user and should be maintainable, dependable and acceptable.

Awel 7aga el software elle hayetla3 dah lazem yewassal w yelabbi el wazayef el matloba wel performance lel user wlazem ykon qabel l2eno yt3melo maintenance w dependable w acceptable

1- Maintainability

- Software must evolve to meet changing needs.

- Software should be written in such a way so that it can evolve to meet the changing needs of customers.

-This is a critical attribute because software change is an inevitable requirement of a changing business environment.

- el software lazem yettawar 3shan yelabbi e7tyagat el ta3'yeer el byetteleb
- el software lazem yetkteb b taree2a tesa3do eno ye2dar yettawar ba3d kda w yet3'ayar 3la 7asab ma el customer 3ayz
- w de to3tabar sema 7asema aw asaseya y3ni .. 3shan ta3'yeer el software hwa shart la mafar menno f bee2et el a3mal el mota3'ayera .. y3ni el business environment byet3'ayar fa tabi3i el software kman yet3'ayar ..

2- Dependability and security

- Software must be trustworthy.

- Software dependability includes a range of characteristics including reliability, security and safety.

- Dependable software should not cause physical or economic damage in the event of system failure.

- Malicious users should not be able to access or damage the system

- el software lazem ykon gadeer bel seqa

- gadaret el software (software dependability) .. btetkawen mn magmo3a mn el 5sa2es heya de elle bet5aleeh gadeer .. zay :

reliability : mawsooqeya .. aw en el software dah ykon mawsooq feeh (gadir bel seqa)

security : en el software ykon secured w 2amaan

safety : nafs ma3na el security bas zay mat2ol en el secutiry y3ni el virus wel safety 3la el human nafso

- el software el gadeer dah mayenfa3sh yesabeb ay tadmeer fezya2i aw eqtesadi lma ye7sal failure lel system .. y3ni msh ma3na en system ye7sallo failure en ye7sal tadmeer fezyaa2i zay aslaak tet7ere2 aw chipat tet7ere2 ... wla tadmeer eqtsadi en y3ni el gehaz yekalefak floos gamda lel taslee7 maho kda fakes menno a7sn ..

- fe users esmohom malicious users .. dol users elle bye5rebo el systems zay el hackers kda .. bey2ollak b2a 3shan el software dah ykon GOOD laezm el malicious users dol maye2darosh y5osho aw ye3melo access aw yedammaro el system dah

3- Efficiency

- Software should not make wasteful use of system resources; Efficiency therefore includes responsiveness, processing time, memory utilisation, etc.

- el software dah msh el mafrood enno yesta5dem goz2 mn el system resourses bdon lazma .. y3ni maykonsh hwa me7tag processor so3'ayar w ya5od kol sor3et el processor 3al fadi .. da kda estehlaak wlaw msh bye7sal yb2a software GOOD .. (y3ni el efficiency eno msh byestahlek el resources) .. w 3shan kda el efficiency btetkawen men

  responsiveness : el estegaba

  processing time : wa2t 3amal el process

  memory utilisation : ad eh el esta5demo mn el memory

  y3ni mn el a5er el efficiency en el software mayestahleksh system resources 3al fadi zay enno ya5od wa2t kbeer fel estegaba lma y3ml run msln .. aw lma ye3mel processing ya5od wa2t kbeer 3al fadi .. aw ykon mesa7to so3'ayara bas wa5ed goz2 kbeer mn l memory wa ha ka za ..

4- Acceptability

- Software must accepted by the users for which it was designed. This means it must be understandable, usable and compatible with other systems that they use.

- el software lazem ykon ma2bool mn el user elle et3amal design lel bernameg 3shano .. w da ma3nah en el software dah lazem ykon :

  understandable : mafhoom

  usable : qabel lel este5dam (sahl y3ni)

  compatible with other systems that user use : y3ni eno yetwafe2 ma3 ay system tani el user yet3amel m3ah .. zay kda windows 7 lazem ye3raf yet3amel ma3 winows 8

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Lec 2 : Software Processes

Process model

- Waterfall model
- Evolutionary development (Iterative Development) Throw-away, exploratory, Aglie method [XP]
- Component Based Software Engineering

software process model

- **Is an abstract representation of a process.**
- **It presents a description of a process from some particular perspective**

el model bta3 el software process:

- hwa 3bara 3an tamseel tagreebi lel process.
- bey2addem wasf lel process mn manzor mo3ayan

y3ni mn l a5er el model da 3bara 3an eno beymassel kol process weybayen mowasafatha mn manzor mo3ayan

Kol project byet3emel wel nas bteshta3'al feeh .. byeb2a leeh system mo3ayan .. el system dah lazem nemshi 3aleh b taree2a sabta 3shan n5allas sho3'l fel project dah … wna hena 3andi 3 toro2 bnemshi 3alehom 3shan nesht3'al 3la el project dah aw el system dah …
W hya de elle esmaha el generic software process models..

Generic Software Process Models

1) The waterfall model

• **Separate and distinct phases of specification and development**

awel model 3bara 3an mara7el (phases) .. monfasela w motamayeza .. mn mwasafat w developments ..

y3ni basht3'al fel project 3an taree2 mara7el bamshi 3aleha step by step .. wel mara7el de kol wa7da feha monfasela 3an le Tanya .. w motamayeza .. wel mara7el de feeha el mwasafat bta3et el project w feha el developments el ha3melha 3aleh..

2) Evolutionary development (Iterative Development)

• **Specification and development are interleaved**

tani model mn esmo tatweer el tatweer .. maho evolutionary y3ni tatweeri .. w development y3ni tatweer .. fa hwa esmo tatweer tatweeri .. evolutionary depelopment .. aw iterative development .. y3ni tatweer motakarer ..

bey2ollak b2a en el model dah metgamma3 feeh el specification wel development interleaved .. interleaved y3ni da5leen f ba3d ..

3la 3aks el waterfall .. henak kanet met2asema mara7el wkol mar7ala malhash da3wa bel Tanya .. laken hena kollo da5el f ba3do (interleaved) .. wbardo zay el waterfall metkawen mn el specification bta3et el project wel development el haytem 3aleh

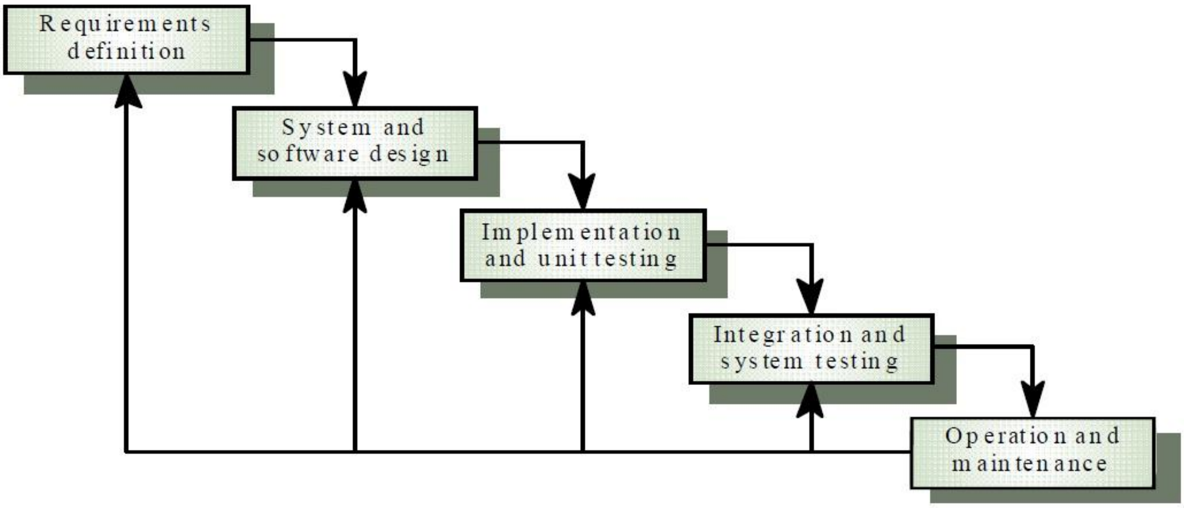
3) Component Based Software Engineering (CBSE)

• **The system is assembled from existing components**

talet model whwa 3bara 3an handaset el software btaree2a mo3tamed 3la el mokawenat ..

wdah beytem tagme3 el system feeh mn mokawenaat mawgooda already y3ni y3ni msh zay el 2 models el fato el kano byesta5demo specification w development .. la2 ana hena hagamam3 el system kollo mn el components el already existed

- **The Waterfall Model**



Waterfall model phases

- **Requirements analysis and definition** // bn3ml ta7leel w ta3reef lel requirements
- **System and software design** // bnebda2 nesamem el system wel software
- **Implementation and unit testing** // bnebda2 ne3mel test lel implementation + unit
- **Integration and system testing** // bnebda2 ne3mel test lel integration + system
- **Operation and maintenance** // ba3d man5allas ne3mel 3maleyat w seyanat

The main drawback of the waterfall model is the difficulty of accommodating change after the process is underway. One phase has to be complete before moving onto the next phase.

drawback = 3eeb

el 3eeb el ra2eesi lel waterfall model .. heya so3obet estee3ab el ta3'yeer ba3d matkon el process sh3'ala .. y3ni kol mar7ala lazem te5las abl may5osh fel mar7ala el gdeeda

y3ni bey2ollak en 3eb el waterfall model eno law wa2ef 3and phase aw mar7ala mo3ayana .. msh bye2dar ye3mel ta3'yeer le mar7ala tanya .. byestanna el awel ye5allas sho3'l fel phase de wba3d kda yb2a y3mel el ta3’yeer el matlob

Waterfall model problems

 Inflexible partitioning process.

3maleyet el ta2seem (partitioning) bteb2a msh marena

 Model is mostly used for large systems of the project into distinct stages
 This makes it difficult to respond to changing customer requirements

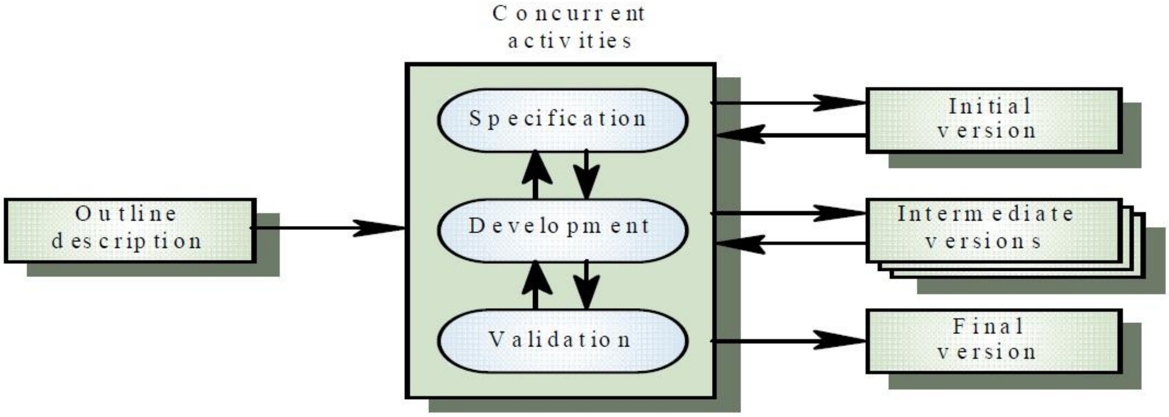
3'aleban el model dah byosta5dam fel large systems bta3et el project lmara7el motamayeza .. y3ni badal ma hwa system kbeer bey7awelo le mara7el kol mar7ala momayaza .. w da tb3n yo3tabar 3eb .. l2en da beysa33ab el estegaba lel le talabat el customers el mota3'ayera ..

3shan zay ma olna el waterfall msh btestaw3eb ta3'yeer ye7sal .. wel sabab en el waterfall btemsek system kamel wet2asemo le distict stages .. w kol phase aw stage lma bykon feha process mabyenfa3sh ye7sal ta3'yeer 3'er lma l process de te5las .. fa bel tali byeb2a sa3b awi lma system kbeer yet2asem le distinct stages enno ye7sal ta3'yeer aw ye7sal respond lel changing customer requirements

 Therefore, this model is only appropriate when the requirements are well understood and changes will be fairly limited during the design engineering

 3shan kda el model dah bykoon monaseb lma el motatalebat btkon mafhoma kwayes gedan wel ta3'yeerat hatkoon ma7dooda gedan 5elal 3maleyet el design engineering

- **Evolutionary development (Iterative Development)**



→**It is based on the idea of developing an initial implementation, exposing this to user comments and refining it through many versions until an adequate system has been developed.**

el evolutionary develpment model bey2om 3la fekret developing an initial implementation .. y3ni zay mat2ol kda code awalii .. wye3red el initial implementation dah lel user wyshof commentat el user 3aleh .. wyebda2 y3mel refining leeh aw ta3deel w tekraar lel implementation dah 3an taree2 kaza version .. l7ad ma yewsal lel system el monaseb ...

y3ni zay ay bernameg kda byenzel .. byebda2 version adeem .. el version dah et3amal 3an taree2 "developing an initial implementation" wba3d kda byenzel el soo2 wel nas tbda2 teddi commentat 3an el 7agat el msh 3agbaha fel bernameg .. yebda2o homma yetawaro fel bernameg dah b aktar mn version l7ad mayewsal lel appropriate system yb2a developed ..

w de el 5atawat bta3et el model :

» **Develop “quick and dirty” system quickly;** //ye3mel develop lsystem bsor3a w msh 7lw
» **Expose to user comment;** //ye3redo lel user comment
» **Refine;** //yetawaro w ykararo tani
» **Until appropriate system is developed.** // l7ad maytem 3mal develop le system monaseb

→**Specification, development and validation are interleaved rather than separate, with rapid feedback across activities.**

El mwasafat (specifications) bta3et el system wel development wel validation byeb2o da5leen f ba3d (interleaved) a7sn mn enohom ykono separate (monfaselin) .. ma3 feedback saree3 3an taree2 el activities .. el hwa el feedback bta3 el user comments ..

There are two fundamental types of evolutionary development:
Fe no3een men el evolutionary development

1- **Exploratory development** //el development el estekshafy

- **Objective is to work with customers to explore their requirements and deliver a final system from an initial outline specification.**

el hadaf menno hwa el ta3amol ma3 el customers w ekteshaf motatalebathom w ta2deem el system el neha2i men el initial outline specification .. elle hwa mwasafat el mo5atat el 2ola..

- **Should start with well-understood requirements.**

lazem yebda2 whwa fahem kwayes eh heya el motatalebaat

- **The system evolves by adding new features as they are proposed by customer.**

beytem tatweer el system 3an taree2 adding new features aw momayezaat gdeeda lel bernameg el customer eqtara7ha lel bernameg

2- **Throw-away prototyping** //namozag throw-away

- **Objective is to understand the system requirements and hence develop a better requirements definition for the system.**

el hadaf menno hwa fehm motatalebat el system w wad3 motatalebat a7san t3arraf el system ..

y3ni elle fat kan mrakez 3la el customer nafso w motatalebaato .. ama hena mrakez ma3 el system w motatalebato w beyshof eh el motatalebat el a7san elle t3arraf el system bshakl afdal ..

- **The prototype concentrates on experimenting with the customer requirements that are poorly understood.**

el prototype aw el namozag dah beyrakkez 3la tagarob ma3 motatalebat el customers elle btetfhem bshakl msh kwayes ...

y3ni 3shan el prototype yefham el system requirements .. byettar enno yerakez 3la tagarob ma3 el customer requirements .. elle btb2a poorly understood

<u>Advantages</u> //momayezat el evolutionary development model

- **It is more effective than waterfall approach in producing systems that meet the immediate needs of customers.**

El evolutionary development more effective than el waterfall approach (nahg) .. mn na7yet entag system yelabbi el e7teyagat el fawreya lel customers ..

y3ni belnesba le talbeyet el immediate customers needs .. el evolutionary development aktar fa3elya mn el waterfall

- **The specification can be developed incrementally.**

el specification fel evolutionary development momken tettawwar btaree2a tadreegeya

- **Users develop a better understanding of their problem; this can be reflected in the software system.**

el users beytawaro fehm a7san le mshakelhom .. w da byen3ekes f system el software ..

y3ni mn el a5er fel evolutionary development bne2dar nefham moshklet el users b sohola wdah sababo en el software system mesahhel el 7etta de ..

Problems

• **The process is invisible. Managers need regular deliverables to measure progress. If systems are developed quickly, it is not cost–effective to produce documents that reflect every version of the system.**

el moshkela en el process el bte7sal bteb2a 3'er mar2eya .. fa l modereen bye7tago 7aga esmaha "regular deliverables" .. y3ni 7agat dayman met3awedeen tewsallohom .. 3shan y2eeso beha el progress el beytem ..

law el systems bye7salaha develop bsor3a .. msh byeb2a mo2aser mn na7yet el cost enena n3ml produce le documents elle bte3kes kol version mn el system

• **Systems are often poorly structured:**

Continual change → tends to corrupt the software structure

• **Incorporating software changes → becomes increasingly difficult and costly.**

• el systems dayman btetbeni bshakl saye2 .. l2en bye7sal ta3'yeer bshakl mostamer wda bey2addi ela fasaad haykal el software ..

• w damg el ta3'yeerat el bte7sal 3al software beyzeed s3oba w taklefa

tawdee7 lel fat dah : .. mn el a5er y3ni bey2ollak en bema enena kol shwaya nenazel versions mmo5talefa w bno3od n3'ayar kteer fel structure bta3 el software .. da bey2addi l2en el system byeb2a poorly structured .. wel ta3'yeer el kteer dah bey corrupt el software structure

tany 7aga en 3maleyet el incorporating software changes .. whya enena benedmeg ta3'yeerat kteer mn el software el kol mara bne3melha .. zay msln enena nzawed shwayet ta3'yeer 3la version adeem .. kda ana 3amalt incorporating software changes .. el 3amaleya de b2a moshkletha enaha betzeed fel (difficult + costly) (so3oba + flos)

• **Special skills (e.g. in languages for rapid prototyping) may be required**

mn demn mshakelha en lazem yb2a el sha3'aleen feeha 3andohom special skills .. zay maslan fel lo3'aat zay el "rapid prototyping" beykoon matloob

El moqarna de f exam 2010-2011 final

	Waterfall	Evolutionary
Advantages	-Visibility: Gives the managers a clear picture of project status at a given point in time. - Documentation: The system is completely and properly documented - Completeness: If all deliverables/outputs of a phase are complete, then the phase is completed - Versatility: Used to produce almost every kind of software	- It is more effective than waterfall approach in producing systems that meet the immediate needs of customers. - The specification can be developed incrementally. - Users develop a better understanding of their problem; this can be reflected in the software system.
Problems	The main drawback of the waterfall model is the difficulty of accommodating change after the process is underway. One phase has to be complete before moving onto the next phase.	- The process is invisible. Managers need regular deliverables to measure progress. If systems are developed quickly, it is not cost – effective to produce documents that reflect every version of the system. - Systems are often poorly structured. Continual change tends to corrupt the software structure and incorporating software changes becomes increasingly difficult and costly. - Special skills (e.g. in languages for rapid prototyping) may be required
Applicability	- this model is only appropriate when the requirements are well-understood and changes will be fairly limited during the design engineering - The waterfall model is mostly used for large system engineering projects where a system is developed at several sites	-For small or medium-size interactive systems - For parts of large systems (e.g. the user interface) - For short-lifetime systems - Particularly suitable where:detailed requirements are not possible; powerful development tools (e.g. GUI) are available

Shar7 el moqarna de:

	Waterfall	Evolutionary
Advantages	• byeb2a visible wbtedi el managers sora wad7a lel status bta3 el project f no2ta mo3ayana fwa2t mo3ayan • kol 7aga fel system btetgamma3 f document • law kol el outputs bta3et mar7ala mn el mara7el 5elset yb2a el mar7ala de 5elset • versatility = mota3adedet el gawaneb .. bma3na enaha btosta5dam 3shan ne3mel beha kol anwa3 el software	• mo2asera aktar mn el waterfall f entag systems mashie ma3 e7tyagat el customers el fawrya • el specifications bta3etha momken tettawwar bsora tadreegeya • el users beytawaro toro2 a7sn yebayeno beha el homa fahmeeno wdah byen3ekes 3la el software system b2eno byeb2a fahem e7tyagat el customers
Problems	el 3eeb el ra2eesi lel waterfall model .. heya so3obet estee3ab el ta3'yeer ba3d matkon el process sh3'ala .. y3ni kol mar7ala lazem te5las abl may5osh fel mar7ala el gdeeda	A5er 3youb etshara7et fel evolutionary last page (7efzz :D)
Applicability	• El model dah monaseb JUST lma tkon kol el requirements mafhoma gedan wel ta3'yeerat feha ma7doda gedan 5elal 3maleyet el design engineering • El waterfall model 3'aleban byosta5dam lel large system bta3et projectat el handasa welle feha el system byet3mello develop f aktar mn site	• Lel systems el motafa3la el so3'ayara aw el motawaseta fel 7agm • Le agza2 mn el large system (zay el user interface) elle hwa ba3d ma l project ye5las el model dah yemsek el interface bta3o • Lel systems elle wa2taha olayel • 5assatan tkon monasba .. lma el motatalebat el tafseeleya (detailed requirements) tkon msh momkena .. w adawat el tatweer el qaweya zay el (GUI bta3 el interface) tkon mawgooda

(final exam 2010-2011)

Explain how a mixed model (Waterfall + Evolutionary) can be used in a project.

Wadda7 ezay ne2dar ne3mel model mix ben el etneen models dol wnesta5demo f project ?

For large system , it is better to use a mixed process between waterfall and evolutionary.

How?

-We develop a system using throw-away prototyping to resolve the uncertainties in the system specification.

- then re-implement using waterfall for a parts of the system that are well understood

- and other parts (user interface) should be developed using the exploratory approach.

Belnesba le large system .. mn el afdal enena nesta5dem el mixed process mabeen el waterfall wel evolutionary .. ezay ?? ..

e7na hane3mel develop lel system 3an taree2 el throw-away prototyping .. we7na olna mn shwaya en el prototype dah byehtam bel system requirements wbyefhamha weytawwarha ..

fa hena bey2ollak hanesta5dem el thow-away prototype de 3shan n7el el 7agat elle msh wasqeen menha f mawasafat el system ..

y3ni law el system leh 7agat mo3ayana msh wad7a (msh mota2akedeen menha) .. fel specifications ba3et el system dah (uncertainties in system specification) .. el prototype dah bema eno byet3amel ma3 el system requirements aslnn .. hwa el haye2dar ye7el el 7agat elle msh wasqeen menha fel specifications bta3et el system

ba3d maytem fehm el 7agat de 3an tari2 el prototype dah .. ne2dar sa3etha nesta5dem el waterfall .. bne3mel e3adet tanfeez lel agza2 mn el system elle ba2et 5laas mafhooma (3shan olna abl kda en el waterfall byet3amel ma3 el 7agat el mafhoma gedan w el ta3'yeer feha ma7dod gedan)

w fe agzaa2 fel system bnesta5dem feeha el Exploratory Approach (no3 mn anwa3 el evolutionary model) w da mas2ool 3an agza2 el user interface yb2a 5olaset el kalam elle fat dah .. eni law odami system w 3ayz astafeed bel 2 models dol ma3 ba3d

el waterfall model bya5od el 7agat elle already ba2et mafhoma wmsh haye7sal feha ta3'yeer wyesht3'al feha fel large systems
amma el evolutionary feeh goz2een .. goz2 el exploratory wdah 5aas bel user interface w goz2 el throw-away prototype wdah mas2ol 3an enno yewada7 el 7agat elle msh mafhoma abl maya5odha el waterfall wyesht3'al 3aleha

belnesba le tani no3 mn el software processing elle esmo evolutionary leh goz2 ba2i f lecture 6 .. elle byetkalem 3an el incremental development .. yosta7san mn dlwa2ti troo7 tazko mn ta7t lecture 6 (page 40)

Talet noo3 mn el software processing models esmo :

- **Component Based Software Engineering (Reuse-oriented development)**

  Awel no2teteen homa aham no2teteen .. el etneen el tanyeen tawdee7 aktar shwaya (bs fakes)

  – **This approach is based on the existence of a significant number of reusable components.**

    el nahg dah aw el model de bteb2a ayma 3la wgood 3adad kbeer mn el mokawenat el betkon abla l2enaha yet3ad este5damha .. y3ni law 7asal wla2ena system feeh components kteer yenfa3 enaha yet3ad este5damha .. basta5dem el model dah

  – **The system development process focuses on integrating these components into a system rather than developing them from scratch.**

    el system development process beyrakez 3la damg el components de lel system a7sn mn enohom ye3melolha developing mn el sefr .. y3ni el components el olna 3aleha reusable de bye7salaha damg ma3 el system a7sn mn enena n3ml develop lkol component lwa7do

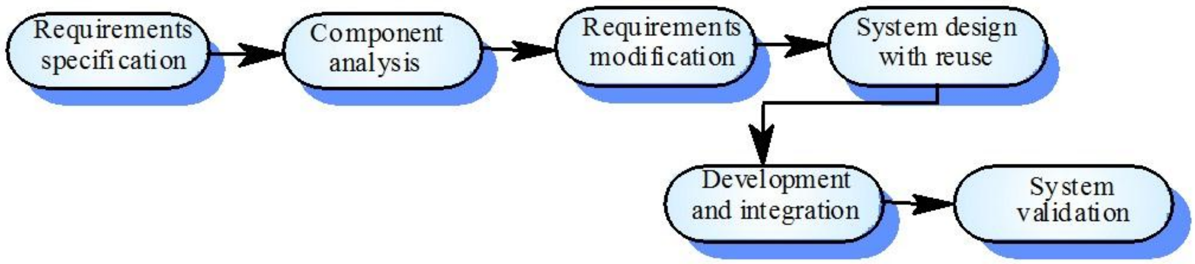
  ❖ **Based on systematic reuse where systems are integrated from existing components or COTS (Commercial-off-the-shelf) systems.**

    bey2om 3la asas menhageyet e3adet este5dam .. elle beytem feha damg el systems ma3 el components aw 7aga esmaha COTS .

❖ **This reuse oriented approach relies on a large base of reusable components and some integrating framework for these components.**

nahg el reuse oriented dah byb2a mowagah na7w qa3eda kbeera mn el components el 2abla le e3adet el este5dam .. w ba3d el 2etar el takamoli (integrating framework) lel components de

Draw a chart that illustrates the <u>Component Based Software Engineering development</u> process?



Lec 3: Software Requirements

- ➢ Functional req.
- ➢ Non-functional req.

- Definitions
- Examples
- Req. measures
- Tree
- Type of req.

(lel fehm)

System Requirements

• Requirements are the descriptions of the system services and its operational constraints that are generated during the requirements engineering process.

Requirements engineering

• The process of establishing the services that the customer requires from a system and the
constraints under which it operates and is developed.

- **Types of requirement**

  1. **User requirements**

  - User requirements are high-level statements of the services the system provides and its operational constraints. User requirements should be written using natural language, tables and diagrams. Written for customers.

  - el user requirements da 3bara 3an mostawa 3ali mn bayanat btet3emel 3shan el 5adamat elle beywafarha el system w qyood el 3maleyat elle bettem de .. y3ni 5adamat + el qyod el 3maleyat bta3et el 5adamat de (services + its operational consraints)

  - lma el user bye7tag motatalebat lazem yektebha lel customers beste5dam:
  1) lo3'et el tabi3a 2) gadawel 3) rsom byanya

  y3ni el 5olasa .. el user requirements de 3bara 3an talabat bey2ademha el user lel customers 3shan me7tageen services mn el system .. w fel ta3reef el talabat de 3bara 3an (high-level statements) .. 3shan ya5do 5edma el system beywaffarha w kman by3rafo eh hya el qyod el byemshi 3aleha el system whwab ye3mel el 5adamat de

2. **• System requirements**

- A structured document setting out detailed descriptions of the system's functions, services and operational constraints. Defines what should be implemented so may be part of a contract between client and contractor.

- We use natural language to write the system requirements. However, because system requirements are more detailed than user requirements, natural language specifications can be confusing and hard to understand.

el system requirements aw motatalebat el system

de 3bara 3an document monazama .. bet7aded wasf tafseeli le (wazayef + 5adamat + qyod el 3maleyat) elle bye3melha el system nafso ..

el document de bet3arraf eh elle el mafrod yetnafez .. 3shan kda momken n2ol enno goz2 mn el 3a2d mabeen el 3ameel wel m2awel (elle hwa bye3mel el 3a2d nafso)

men el a5er el system requirement 3bara 3an document monazam feeh descriptions 3an wazayef el system wel 5adamat el ye2dar ye3melha el system wel qyod el byemshi 3aleha el system fel 3maleyat el bye3melha ..

y3ni ana law client fa ba5osh fel sherka ashof el system requirements .. bashoof el detailed description .. w da ba7aded en kan el 3agebni el (functions wel services) bta3et el system da wla la2 .. law 3agabetni babda2 a3mel 3a2d ma3 el contractor (el m2awel aw el ragel el bye3mel m3aya el 3a2d) ..

fa 3shan kda ana ba2ool en el system requirement de to3tabar goz2 mn el 3a2d mabeen el client wel contractor ...

- bnesta5dem natural language bardo 3shan nekteb el system requirements .. wama3azalek, 3shan el system requirements bteb2a feha tafaseel aktar mn el user requirements .. fa mwasafat el natural language momken tkoon sa3b fehmaha

The main advantages of using a standard format to specify requirements?
el meeza el asaseya f2eni asta5dem shakl sabet 3shan a7aded el requirements ?

1) ***All requirements have the same format so are easier to read.***

2) ***The definition of form fields mean that writers are less likely to forget to include information.***

3) ***Some automated processing is possible.***

1) kol el requirements leha nafs el format fa byeb2a sahla enena ne2raha

2) ta3reef el fields bta3et el form ma3naha en el byektebo el requirements de a2al fel ta3arod l2enohom yenso enohom y7oto kol el information

ma7na law 3malna standard format hatkon kol 7aga sabta fmakanha w met3arrafa fa el byekteb el requirements msh hayensa y7ot information

3) momken ye7sal shwayet 3maleyat automatic w de 7aga 7lwa.

(el dr. 2alo)

El etneen dol lel ta3reef bas (define)

Functional requirements //motatalebat wazeefya

Statements of services the system should provide how the system should react to particular inputs and how the system should behave in particular situations.

de 3bara 3an bayanat 3an el 5adamat bta3et el system elle bet2ol:
- ezay el system hayatafa3al ma3 inputs mo3ayana
- ezay el system hayet3amel f mawaqef mo3ayana

Non-functional requirements //motatalebat 3ér wazifya :D

Constraints on the services or functions offered by the system such as timing constraints, constraints on the development process, standards, etc.

de 3bara 3an qyood aw sawabet 3la el 5adamat aw el wazayef bta3et el system wlma n2ool qoyood .. ma3naha el 7agat elle bto7kom el system lma byeegi ywaffar 5edma .. shoroot lazem yemshi 3aleha el system lma ye3mel el services de

zay:

timing constraints : el wa2t elle haye3mel feh el service aw el function

constraints on the development process : lma y3mel tatweer le mashroo3 masalan lazem yemshi 3la shroot t7a2a2 talabat el tatweer dah msh ay tatweer w5las

standards : ma3ayeer aw 7agat sabta lazem yemshi 3aleha el system whwa sh3'al

Hena kalam aktar 3anhom msh bas ta3reef

- **Functional requirements**

 Describe functionality or system services.

el functional requirements btewsef (wazayef AW 5adamat) el system

 Depend on the type of software, expected users and the type of system where the software is used.

el functional requirements bte3temed 3la :
1) no3 el software
2) no3 el system elle el software byosta5dam feeh
3) el users el motawaqa3een

 Problems arise when requirements are not precisely stated.

bye7sal mshakel lma el requirements matkonsh mazkoora b deqqa

 Ambiguous requirements may be interpreted in different ways by developers and users.

ambiguous = مبهم \ غامض

el requirements elle msh wad7a (mobhama) momken tettargem btoro2 mo5talefa 3an taree2 el developers wel users ..

 The functional requirements of a system should be precise, complete and consistent.

el functional requirements bta3et el syste lazem tkon:
1) daqeeqa
2) kamla
3) motanasqa

 Completeness means that all services required by the user should be defined.

belnesba lel completeness bta3 el requirements ma3nah en kol el services el matloba mn el user lazem tkoon met3arrafa

 Consistency means that requirements should not have contradictory definitions.

contradictory = متناقض
belnesba lel consistency bta3 el requirements ma3naha enaha mayenfa3sh tkon feha ta3reefat motanaqda

- ## **Non-functional requirements**

Classifications //tasneef (anwaa3o)

1) **Product requirements** //motatalebat el product

   Requirements which specify that the delivered product must behave in a particular way

   e.g: execution speed, reliability, etc.

   el no3 dah mn el requirements bey7aded en el product elle wesel lazem yetsarraf btari2a mo3ayana ..
   example: sor3et el tanfeez - el mawthooqya aw en el product ykon mawsooq feeh

   y3ni mn el a5er kda .. lma beykon 5las el product wesel w haybda2 ye3mel run .. el no3 dah mn el requirements bey7aded shwayet tasarofat lazem el product lma ye5las ye3melha .. zay sor3et el tanfeez .. shwayet talabat bet2ol ana 3ayez sor3et tanfez el product dah tb2a kaza .. w de to3tabar contraints zay ma olna .. w zay msln el reliability .. y3ni talabat bet2ol lazem el product dah ykon mawsooq feeh ..

   wbkda nefham en el product requirements de 3bara 3an ta7deed lel "behave" aw el tasarof bta3 el product dah ..

2) **Organisational requirements** //motatalebat tanzeemeya

   Requirements which are a consequence of organisational policies and procedures
   e.g. process standards used, implementation requirements, etc.

   consequence : نتيجة ل
   policies : سياسات
   procedures : إجرائات

   w de shwayet requirements btet3emel nateega lshwayet seyasat w egra2aat tanzeemeya

   y3ni lama ykon fe monazama 3ayza te3mel system mashi 3la asas homma 3ayzeeno .. mashi 3la seyasaat w egra2aat homma 3ayzenha (organisational policies and procedures) .. sa3etha byesta5demo el organisational requirements yotlobo beeha el 7agat elle homma 3ayzenha de..

   examples:

   process standards : ma3ayeer homma 3ayzeen yemsho 3aleha fel processing

   implementation requirements : byotlobo shoroot lel tanfeez

3) **<u>External requirements</u>** //motatalebat 5aregya

Requirements which arise from factors which are external to the system and its development process e.g. interoperability requirements, legislative requirements, etc.

motatalebat bte7sal bsabab 3awamel 5aregeya 3an el system w 3amaleyet tatweero ..

bma3na : 3adatan lma beytem ta2deem motatalebat lel system .. beykon el system hwa el mas2ol 3anha ..
laken el requirements hena 7aga la el system leeh da3wa beeha .. wla el development process el bte7sal gwa el system leha da3wa beeha ..

 Non-functional requirements may be more critical than functional requirements. If these are not met, the system is useless

el non functional requirements aktar ahameya mn el functional requirements .. w da law matammesh el system yb2a malosh fayda

 Conflicts between different non-functional requirements are common in complex systems.

F 7alet el systems el mo3aqada .. el neza3at ben el non-functional requirements bteb2a sha2e3a

(exam 2011-2012)

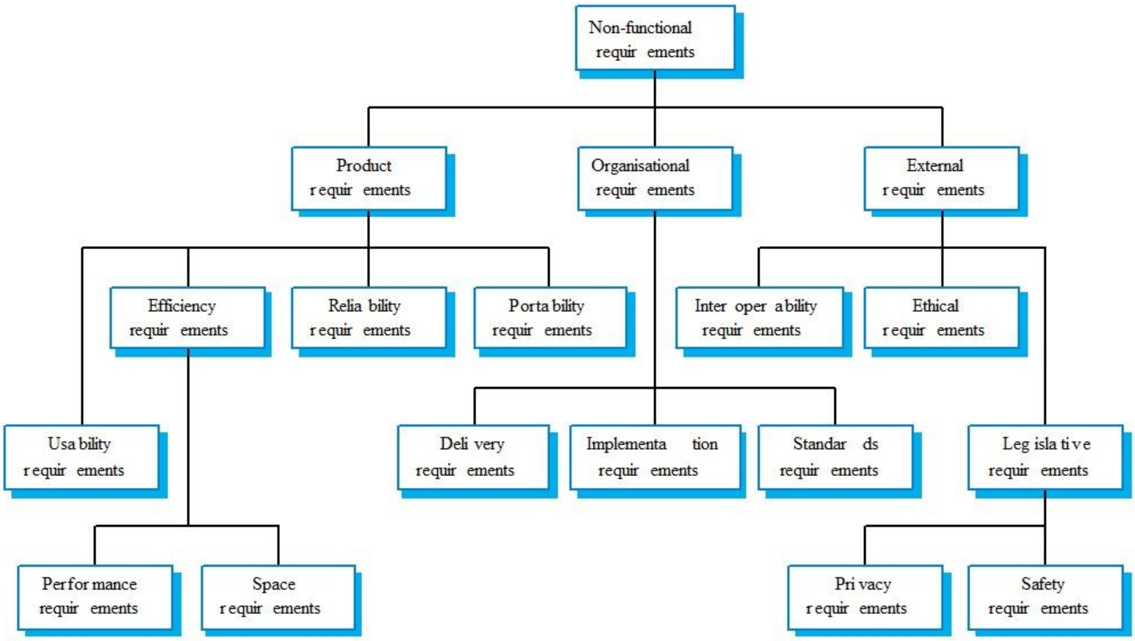
- **Requirements measures**

Kol 5aseya byeb2a leeha we7det qeyas ben2ees beeha el 7aga de

Property	Measure
Speed Ba2is el sor3a b	• **Processed transactions/second** // 3dad el 3amaleyat elle 5elset fel sanya • **User/Event response time** // zaman estegabet el user / el 7adas • **Screen refresh time** // el wa2t el byet3mle feh refresh lel shasha
Size Ba2is el size b	• **Size M Bytes** //el size byet2as bel Bytes • **Number of ROM chips** // 3adad el chips fel ROM
Ease of use 3shan a3raf sholet el este5dam	• **Training time** //3shan a3raf sholet el este5dam a3raf wa2t el training • **Number of help frames** // 3dad el frames bta3et el help
Reliability 3shan a3raf ad eh el bta3 dah mawsooq feeh	• **Time to restart after failure** // el wa2t el bya5do 3shan y3ml restart lma ye7sal fail • **Percentage of events causing failure** // nesbet el events el bte3mel fail • **Probability of data corruption on failure** // e7tmalyet talaf el data 3al fashal !
Robustness 3shan a3raf ad eh matanet el system	• **Time to restart after failure /** • **Percentage of events causing failure** • **Probability of data corruption on failure** //nafs el fato .. yb2a el 3 dol bey7adedo el seqa fel system w qowet w matanet el system (Reliability + Robustness)
Portability 3shan a3raf e7tmaleyet aw kafa2et el system	• **Percentage of target dependent statements** // nesbet el bayanat el mo3tameda 3la el hadaf bta3 el product .. y3ni hal el statements leha 3laqa bel target wla la2 • **Number of target systems** // 3adad el systems elle weslet lel target el matlob

(midterm 2012-2013)

Draw a diagram that shows the classifications tree of non-functional requirements



Lec 4 (Use case & Class Diagram)

Matlob feha 7agteen .. awalan ne7faz shwayet ta3refat saneyan n3raf n7el el mas2ala

El goz2 el 3amaly videos fehm ezay tet7al el mas2ala

El goz2 el nazary:

The UML

- **is a standard representation devised by the developers of widely used object-oriented analysis and design methods.**
- **It has become an effective standard for object-oriented modelling.**

Belnesba lel Use case wel Class diagram .. hakteb el nazary bta3hom (mn lecture 4) ba3d mafham el video 3shan el video byeshra7 el kalam el nazari

Yb2a ba3d el fehm:

1- Akteb el nazari bta3 page 4 flecture 4 (use case + class diagram)
2- Agamma3 as2elet el use case wel class diagram mn el exams

Lecture 5 : Architectural Design

(el gai lel fehm .. y3ni el dr. m2alsh 3aleh mohem bas efham da 3shan tefham el mohem)

Hanebda2 fel lecture de netkalem 3an 7aga gdeeda 5ales esmaha el architectural design ..

Definitions

 The software architecture of a program or computing system is the structure or structures of the system, which comprise software components, the externally visible properties of those components, and the relationships between them.

Eh hwa el software architecture ??

el software architecture bta3 program aw bta3 computing system .. 3bara 3an haykal aw benaa2 lel system .. aw magmo3a mn el structures lel system dah .. wel software architecture dah byetkawen mn (comprise) :
1) mokawenat el software nafso (software components)
2) 5asa2es mar2ya mn el 5areg lel mokawenat de ..(externally visiple properties)
w mar2ya mn el 5areg hena ma3naha enena n2dar nshof el 5awaas de

3) el 3elaqa benhom

yb2a mn el a5er law bat3amel ma3 program .. el software architecture lel program dah 3bara 3an el haykal bta3 el program el ana sh3'al 3aleh .. wel haykal dah byetkawen mn 3 7agat .. (1) mokawenat el program .. (2) 5sa2es el mokawenat de .. (3) el 3elaqa ben el mokawenat w 5asa2esha

 Architectural Design (AD) is an early stage of the system design process.

eh hwa el architectural design (AD) ?

hwa 3bara 3an mar7ala mobakera mn 3amalyet el system design .. y3ni abl ma ba3mel design lel system ba3mel 7aga esmaha architectural design ..

<u>The distinction from detailed design</u>

Software architecture is the realization of (casts) non-functional requirements and partitions functional requirements, while software design is the realization of functional requirements.

hanmayez been el **<u>Software Architecture</u>** & **<u>Software design</u>**

el **<u>software architecture</u>** 3bara 3an ta72ee2 lel <u>non-functional requirements</u> w agza2 mn functional requirements

amma el **<u>software design</u>** bey7a22a2 el <u>functional requirements</u>

 Software architecture, also described as strategic design, is an activity concerned with global design constraints, such as programming paradigms, architectural styles, component-based software engineering standards, design principles, and law-governed regularities. Detailed design, also described as tactical design, is an activity concerned with local design constraints, such as design patterns, architectural patterns, programming idioms, and refactorings.

 AD is aiming at identifying the sub-systems making up a system and the framework for sub-system control and communication.

architecture design hadafo :

1) Ta7deed el systems el far3ya elle betshakkel system kamel
2) Ta7deed el framework lelta7akom wel etesal bta3 el system el far3i

 Architectural design is a creative process so the process differs depending on the type of system being developed.

el AD 3bara 3an 3amaleya mobtakara wbeltali el 3maleya bte5telef 7asab no3 el system elle bye7sallo develop ..

 The architectural design is normally expressed as a block diagram presenting an overview of the system structure.

 More specific models showing how sub-systems share data, are distributed and interface with each other may also be developed.

fe models mota5asesa aktar betbayen ezay el systems el motafare3a bte3mel share lel data .. w de beytem tawzee3 el models de tafa3olha ma3 ba3d

- ## <u>Architectural design process</u>

□ System structuring
The system is decomposed into several principal sub-systems and communications between these sub-systems are identified

□ Control modelling
A model of the control relationships between the different parts of the system is Established

□ Modular decomposition
The identified sub-systems are decomposed into modules

- ## <u>Architectural models</u>

□ Different architectural models may be produced during the design process
□ Static structural model that shows the major system components.
□ Dynamic process model that shows the process structure of the system.
□ Interface model that defines sub-system interfaces.
□ Relationships model such as a data-flow model that shows sub-system relationships.
□ Distribution model that shows how sub-systems are distributed across computers.

@ (kol el gay el dr. 2al mohem)

- **System organization**

- **Reflects the basic strategy that is used to structure a system.**

  el system organization bye3kes el estrateejya el asaseya elle btosta5dam fe bena2 ay system .. bma3na en el system organization da bey2ol taree2et bena2 el system

- **Three organizational styles are widely used:**
  3 stylat tanzeemeya btosta5dam

  1. **A shared data repository style.**

     awel style esmo "shared data repository"..
     y3ni el data f mkan aw mostawda3 "repository" wel mkan dah byet3melo share

  2. **A shared services and servers style.**

     tani style share le services w servers

  3. **An abstract machine or layered style.**

     style esmo abstract machine aw layered ..
     wlesa han2ol tafaseel 3anno odam

(exam 2011-2012)

- **Describe the Shared data repository architectural model?**

- # The repository model

- **Sub-systems must exchange data. This may be done in two ways:**

  dlwa2ti fe 7aga esmaha sub-systems .. w hya 3ebara 3an enena badal manet3amel ma3 system wa7ed la2 e7na bnet3amel ma3 magmo3et sub-systems byetnafezo ma3 ba3d 3shan newsal lel system el kbeer dah

  fa 3shan dah yetem .. lazem bye7sal estebdal lel data ben el sub-systems .. w da beytem b tare2teen ..

  1) **Shared data is held in a central database or repository and may be accessed by all sub-systems;**

  awel tari2a .. el data yet3melaha share f database markazeya aw f mostawda3 (y3ni database felnos ben kol el sub-systems) .. w dah yenfa3 yet3emel 3aleh access mn kol el sub-systems de

  2) **Each sub-system maintains its own database and passes data explicitly to other sub-systems.**

  tani taree2a kol sub-system lwa7do ye7tefez b database 5assa be w lma ne7tag ne3mel exchange le data ymarar el data de 3latool lel sub-system elle me7tag el data de

- **When large amounts of data are to be shared, the repository model of sharing is most commonly used.**

  lmaa beykoon fe kemmeya kbeera mn el data 3ayzeen ne3melaha share ben el sub-systems .. el repository model lel share hwa el dayman bnesta5demo

- ## Client-server model

 Through the appearance of Local-Area-Networks, PCs came out of their isolation, and were soon not only being connected mutually but also to servers. Client/Server-computing was born.

> ba3d ma zahar mawdo3 el Local-Area-Networks .. wgat aghezet el PC mn 3azlethom .. wkonna orayeb msh bas bne3raf ne3mel connect benna w ben ba3d .. la2 da kman ederna n3ml connect benna w ben server .. hwa dah el wa2t elle etwalad feeh el Client-Server computing ..

 Distributed system model which shows how data and processing is distributed across a range of components.

> el model dah yo3tabar distributed system .. y3ni system metwazza3 .. w dah beybayen ezay el data wel processing byetwaza3o 5elal magmo3a mn el mokawenat ...
> wel tawzee3 hena ma3nah en msln f7warat el network ana msh bat3amel ma3 memory wa7da wla processor wa7da .. la2 dana wazza3t kemeya mn el data wel processing 3an taree2 components kteer el heya el agheza elle moshtareka fel network ..

 Set of stand-alone servers which provide specific services such as printing, data management, etc.

> el model dah byetkawen mn magmo3a mn 7aga esmaha stand-alone servers .. y3ni serverat lwa7daha ma3mol feha share le 7agat bte7tagha ba2eet el agheza .. fa bey2olak hena en el stand-alone servers de betwaffar 5adamat mo3ayana zay el printing wel data wel management .. y3ni server ma3mol feh share le data .. aw server lma ben7eb n3ml print l7aga bnro7 lel server dah 5aas bel tab3 .. aw server tani 5aas bel edara wahakaza .. kol el servers de esmaha stand-alone servers

 Set of clients which call on these services.

> bema en fe magmo3a mn el servers .. fa tb3n fe magmo3a mn el clients elle byestafeedo mn el services elle bte3melha el servers

 Network which allows clients to access servers.

> tb3n lazem ykon fe network 3shan tesma7 lel clients enohom y5osho 3la el servers w yestafeedo beeha...

yb2a mola55as elle fat bey2ool en:

1. el client-server model dah zahar ba3d mazahar el LAN w enena ba2ena n3raf n3ml connect b server

2) w enno 3bara 3an distributed system bey distribute el data wel processing across magmo3a mn el components

3) eno byetkawen mn shwayet 7agat zay:

- magmo3et stand-alone server byeddi 5adamat lel clients (printing-data-management)
- magmo3et clients byestafeedo mn 5admat el servers de
- network tesma7 el clients ennohom ye3mel access 3la el servers de

El 7etta de shar7 abl man5osh fel two w three tier
(N-Tier Architecture)

dlwa2ti hanetkalem 3la 7aga esmaha el n-tier architecture .. dlwa2ti ana lma bagi a3mel msln mobile application .. el application dah byetkawen mn aktar mn tier aw aktar mn layer aw aktar mn goz22 .. wel layers de btet2asem le:

presentation layer : w da el interface elle el user beyshofo 3la el website aw 3la el interface bta3 el application .. el mohem en l layer de gwaha el output elle beyshofo el user

application layer : w da elle beysheel el business logic .. elle hwa el code b2a wel implementation wel programming bta3 el application

data layer : w da el layer elle beysheel el database wbey5azzen kol el data elle bye7tagha el application dah ..

fa kda e7na assemna el application bta3na le 3adad mo3ayan mn el layers aw el tiers .. badal makonna net3amel ma3 el application kollo 3la ba3do loksha wa7da bel interface bta3o bel data bel code bkol 7aga..

wmen fawayed dah en el application yb2a secured .. kol layer lwa7do .. wkman sahl enno yet3emelo maintainance 3shan kol layer leeh mota5ases beta3o yeegi yesalla7o .. wlaw 7asal wgeh msln yesalla7 7aga fel application layer .. malosh da3wa bel presentation wla leh da3wa bel data .. fa de 7aga bte7mi el application bta3na .. wmsln law 7abena nezawwed data aw n3'ayyar fel data hane3mel kda mn 3'er man2arrab men ba2i el layers aw nbawwaz 7aga fel application .

ayyan kan dlwa2ti homma two tiers wla three tiers .. el mohem enena fhemna e7na leeh ben2assem el application aw bnemshi 3la architecture mo3tamed 3la el ta2seema de .. elle esmo (n-tier architecture)

El two tier wel three tier software architectures dol yo3tabaro anwaaa3 men el client/server model .. y3ni manensaash enena kol dah ta7t 3enwan el client/server model .. wmen anwaa3o aw el toro2 elle bnesta5dem feeha el client/server model .. homma el n-tier architectures dool ..

- ## Two Tier Software Architectures (Fat-client model)

Two tier software architectures were developed in the 1980s .

et3amal fe el tamaneenat

The two tier architecture is intended to improve *usability* by supporting a forms-based, user-friendly interface.

et3amal 3shan ye7assen mn qabeleyet el este5dam .. 3an taree2 da3m el interface elle leeh sefateen (1) forms-based (2) user-friendly ..
y3ni interface mo3tamed 3la shakl form .. w ykon sahl 3la l user este5damo

In this model, the server is only responsible for data management. The software on the client implements the application logic and the interactions with the system user.

fel model dah .. welle el mafrod ben2assem feeh el application le two tiers .. el awal hwa el server .. wel tany hwa el software elle byesta5demo el user ..

el server mas2ol 3an 3amal management lel data ..

wel software hwa elle byet3amel m3ah el client .. wda bye3mel 7agten:
1) beynaffez el application logic
2) byetfa3el ma3 el user nafso

y3ni mn el a5er kda fel ta2seema de eh el 7asal ??

el server b2a shayel el data wbeynazzamha

wel software gwah el application logic elle hwa el bernameg nafso bel source code bta3o ... wfnafs el wa2t el software dah hwa el byetfa3el ma3 el user 3an taree2 en el user byesta5dem el software dah

- ## Three Tier Software Architectures(Thin-client model)

A special type of client/server architecture consisting of three well-defined and separate processes, each running on a different platform:

el three tier software architectures .. da noo3 5aas w momayaz mn el client/server architecture byetkawen mn talat 3amaleyat monfaseleen wmet3arafeen kwayes w kol 3maleya menhom bettem 3la platform mo5talef

1. **The user interface, which runs on the user's computer (the *client).***

   awel wa7ed el user interface w da beyrun 3la el computer bta3 el user aw el client welle beeh byet3amel ma3 el application

2. **The functional modules that actually process data. This *middle tier* runs on**

   a server and is often called the application server.
   Tani wa7ed 7aga esmaha functional modules .. aw we7dat bet2om bel wazayef .. w de elle bte3mel process lel data .. y3ni zay ma olna 3aleha abl kda el source code aw el mkan el beytem feeh 3amal implementation lel program .. w da esmo el middle tier w da beyrun f server .. wel server dah esmo application server

3. A database management system (DBMS) that stores the data required by the middle tier.

   talet tier esmo DBMS .. aw database management system .. w da 3bara 3an mkan shayel el data elle byetlobha el middle tier .. tb3n el tier elle shayel el code wel implementation wel processes el bte7sal 3la el data da bye7tag data mn el tier el talet elle m5azen el data kolaha f database

   yb2a 5olaset el three tier architecture :

   3bara 3an 3 3maleyat ..
   1) interface 3la gehaz el user

   2) (middle tier) application server gwah el functional modules el bye7sal feh process lel data

   3) (DBMS) w da elle shayel el data elle bye7tagha el middle tier f database

(hwa matkalemsh 3an abstract machine bas etkalem 3an layer architecture fa msh 3aref leh 3elaqa b dah wla la2)

- <u>Abstract machine (layered) model</u>

 Used to model the interfacing of sub-systems.

 Organises the system into a set of layers (or abstract machines) each of which provide a set of services.

 Supports the incremental development of sub-systems in different layers. When a layer interface changes, only the adjacent layer is affected.

 However, often artificial to structure systems in this way.

- ## Control styles

 Are concerned with the control flow between sub-systems. Distinct from the system decomposition model.

msh e7na konna olna enena bne7tag n3ml exchange lel data between el sub-systems ?? ..

el control styles b2a de 7aga btehtam bel ta7akom fna2l el data de maben el sub-systems

wmenha no3een

- **<u>Centralised control</u>**

One sub-system has overall responsibility for control and starts and stops other sub-systems.

El no3 el awal .. en wa7ed mn el sub-systems ykon leh mas2oleya kamla f enno yet7akem w yebda2 wyenhi el sub-systems el tanyeen

- **<u>Event-based control</u>**

Each sub-system can respond to externally generated events from other sub-systems or the system's environment.

tani taree2a en kol sub-system lwa7do ye2dar yrod 3la el events elle etwalledet mn barra .. mn sub-system tani .. aw mn bee2et el system nafso..

y3ni mn el a5er kda en kol sub-system yb2a mas2ol 3an nafso w 3an el rad 3ala el 7agat elle gayalo mn barra ...

wlessa hane3raf tafaseel aktar 3anhom homma el etneen

- **Centralised control**

A control sub-system takes responsibility for managing the execution of other sub-systems.

da ta3reef tani leeh .. e5tar el ye3gebak feehom w 2olo
hena bey2ol en el centralised control 3bara 3an sub-system mas2ool 3an edaret el execution aw tanfeez kol sub-system tani..

w Bema enena olna en el centralised control da byet7akkem f ba3t w este2bal el data ben el sub-systems .. w da bye7sal 3an taree2 7agteen .. el 2ola heya el call-return wel Tanya manager model:

❖ **Call-return model**

Top-down subroutine model where control starts at the top of a subroutine hierarchy and moves downwards. Applicable to sequential systems.

El call return model dah 3bara model byemshi ben kol etneen sub-systems byet3amlo ma3 ba3d .. bye7sal benhom call w return (send w receive) wel model dah hwa el byet7akem fel ta3amol el bye7sal ben el two sub-systems

W dah model byemshi btaree2a rooteenya mn fo2 leta7t (top-down subroutine model) .. w el start 3ando byebda2 mn el top bta3 el tadreeg el roteeni dah (subroutine hierarchy) .. wbyefdal yenzel lta7t .. w dah byosta5dam felsystems elle betkon tadreegya asln (sequential sytems)

Y3ni mn el a5er law 3andi sequential system elle byeb2a feh el sub-systems mashya bel tadreeg (hierarchy) .. el model dah btebtedi el start bta3to fel top wyenzel btadreeg roteeni lta7t l7ad may5allas kol el system weycontrol koll el ta3amolat ben el sub-systems

❖ Manager model

Applicable to concurrent systems. One system component controls the stopping, starting and coordination of other system processes. Can be implemented in sequential systems as a case statement.

Amma b2a el manager model fa dabyosta5dam ma3 el systems el motazamna (concurrent) .. y3ni el systems el bto3od fatrat kbeera gedann ..

w wa7d mn mokawenat el model dah byet7akkem f nehayet w bedayet w tanseeq el processes bta3et el systesm el Tanya ..

w bey2llak kman enno momken yt3mello implement aw y3ni nesta5demo ma3 el sequential system bardo ka2enno “case statement” aw y3ni ka2ennaha 7ala shazza

y3ni mn el a5er el manager model feh system components beystop wystart weycoordinate el processes bta3et el system el tani .. wbnesta5demo fel concurrent systems .. 5elset

- ## Event-driven systems

Driven by externally generated events where the timing of the event is out with the control of the sub-systems which process the event.

Da ta3reef b tafseel shwaya bas el ta3reef el adeem 7elw .. hwa hena bas mzawwed en el l kalam dah bye7sal lma wa2t el event bye5las wkda .. fakes el ta3reef elle fo2 7elw

Kan be2yol:

tani taree2a en kol sub-system lwa7do ye2dar yrod 3la el events elle etwalledet mn barra .. mn sub-system tani .. aw mn bee2et el system nafso..

y3ni mn el a5er kda en kol sub-system yb2a mas2ol 3an nafso w 3an el rad 3ala el 7agat elle gayalo mn barra ...

Two principal event-driven models

3andena mabda2een fel event-driven models

❖ **Broadcast models.**

An event is broadcast to all sub-systems. Any sub-system which can handle the event may do so.

El broadcast models 3bara 3an en Lma bye7sal en fe event aw moshkela 3ayza tet7al .. byet3mellaha broadcast (aw eza3a) lkol el sub-systems .. wlaw fe sub-system y2dar yhandle el event dah yebtedi yhandelo

❖ **Interrupt-driven models.**

Used in real-time systems where interrupts are detected by an interrupt handler and passed to some other component for processing.

El interrupt-driven models de btosta5dam fel real-time systems .. dlwa2ti ba3d ma sub-system yqarar en hwa el hayhandle el event .. whwa beyhandelo 7asal m3ah interrupt .. hena bye7sal 7aga esmaha interrupt-driven whya en lma ye7sal ekteshaf le interrupt .. 3an taree2 wa7d esmo “interrupt handler” .. wyebda2 ybasi el event le component tani ybda2 ye3mello processing ..

Yb2a tany 3shan el la5bata .. awel 7aga el event bya5do wa7d mn l sub-systems w yhandelo .. ba3d kda yro7 le “interrupt handler” yedetect en fe interrupt fel event fa ybasi el event le component tani y3mel process

1- Broadcast (meen haya5od el event)
2- Sub-system yhandle el event
3- Interrupt handler yshof aw ydetect el interrupt elle fel event
4- Ypass el event le other component 3shan y3ml process mn gdid

– **Other event driven models include spread sheets and production systems.**

W fe event-driven models Tania byeb2a feha “spread sheets” w “production systems” .. wel spread sheets de zay sheets feha el event btetwazza3 3la el sub-systems wel y3raf yhandel el event byesht3’al .. wel production system dah zay mat2ol kda en msh bas bahandle el event wel interrupt la2 dana kman 3andi system byenteg product kamel

Tayb el fat dah kan ta3reef baseet lel broadcast models wel interrupt driven model..

El 7etta el gaya de nekteb aktar shwaya 3anhom

(exam 2011-2012)

Describe the broadcast model used for control modelling in architectural design?

❖ **Broadcast models.**

An event is broadcast to all sub-systems. Any sub-system which can handle the event may do so.

❖ **Broadcast model**

- **Effective in integrating sub-systems on different computers in a network.**

Mo2aser f7etet eno beykamel el sub-systems el mo5talefa elle f network wa7d .. l2en tb3n byet3eref kol sub-system bye3raf yhandle anhi events

- **Sub-systems register an interest in specific events. When these occur, control is transferred to the sub-system which can handle the event.**

  ba3d kaza marra lma ye7sal broadcast el subsystems betsaggel ehtemamo b events mo3ayana .. ba3d ma eder yhandle el no3 dah mn el events kaza marra fa byetsagel 3anndo eno interest in this specific event .. lma da ye7sal .. sa3etha lma ye7sal broadcast gdeed el control bta3 el 3maleya de bey7awwel el event lel subsystem el bye3raf yhandle el event dah (bema enno metsaggel 3ando enno interest lel no3 dah mn el events)

- **Control policy is not embedded in the event and message handler. Sub-systems decide on events of interest to them.**

  mafeesh syaset el mora2ba 3and el ragel el beyhandle el event wel message .. kol sub-sytem beyqarar el events el hwa interest leha

  y3ni mn el a5er el handler bta3 kol sub-system hwa el bey7aded el event dah hwa interest leh wla la2 wmsh bey5osh fel mawdo3 dah control policy .. y3ni m7adesh byet7akkem fel handler dah .. hwa 7or fel interests bta3to.

- **However, sub-systems don't know if or when an event will be handled.**

  El sub-system msh bye3raf hal asln haye3raf yhandle el event dah wla la2 .. wmsh bye3raf emta haye2dar yhandelha (if + when)

Lecture 6:

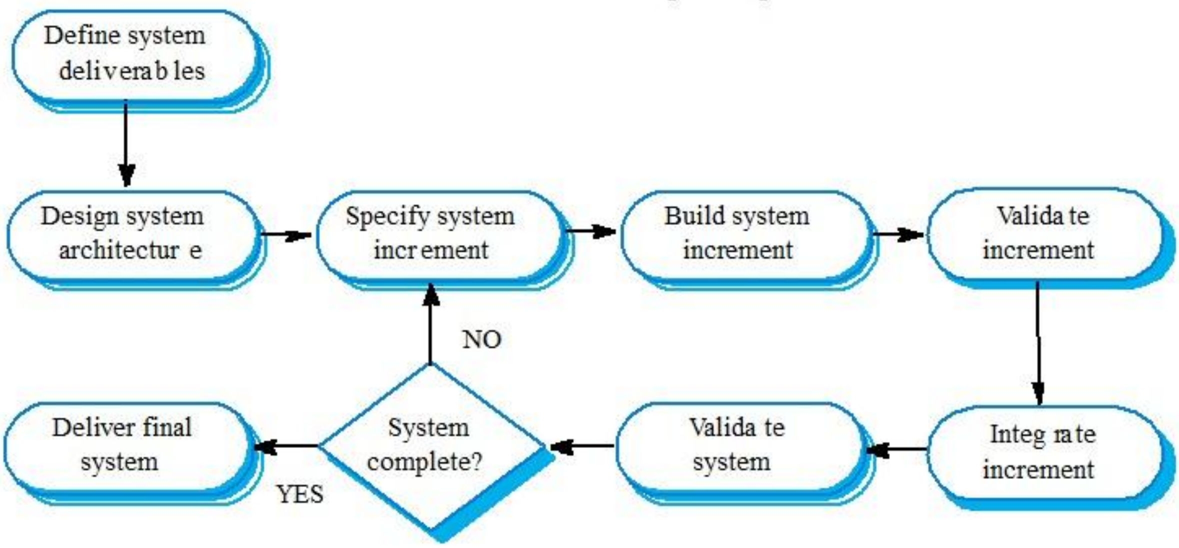
El incremental development dah shwaya 3amel zay el Evolutionary development bta3et lecture 2 .. elle kan feha byet3emel initial implementation wyebda2 yshof comments el users wy3mello integrate lel system w update lel version wytla3 program a7das ..
Wlaw msh mafhom el kam kelma el fato hayetfehmo mn el rasma el gayya..

Hwa el incremental development byetkalem 3an nafs el mawdoo3 ..

Bema enena olna en fel evolutionary el specifications bta3et el system btettawwaar bsora tadreegeya “incremental” .. fa de el rasma el tawdee7ya lel incremental development

(exam 2011-2012)

– **Draw a chart that illustrates the incremental development process.**



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1) Btebtedi b ta3reef el (deliverables / output) bta3 el system (bma3na eh elle 3ayzo yetla3 fel output bta3 el system dah ?)

2) wba3d kda yesamem el arch. Bta3 el system

3) y7aded el increment lel system .. w ta7deed el increment hena ma3naaha (yshof haa feen el moshkela el 3ayzeen netawwarha wnebda2 netawarha tadreegeyan fel laffa elle gaya de)

4) ba3d ma 7adedna el increment nebda2 nebni el increment bta3 el system

5) ye3mel validate lel increment dah 3shan nedman enno 5las el increment etzabat

6) bema en 5las kda validate w damanna eno b2a increment mazboot .. n3mel integrate lel increment dah (y3ni damgo ma3 el system)

7) wba3d kda ye3mel validate lel system nafso ba3d ma b2a (system + increment)

8) w yes2al hal el system kda b2a complete ?? .. law la2 yb2a yerga3 tani lel step bta3et ta7deed el increment lel system mn gdeed .. law ah el system complete .. sa3etha yetalla3 el output (aw ye deliver el final system) ..

Advantages of incremental development

momayezto

• **Accelerated delivery of customer services. Each increment delivers the highest priority functionality to the customer.**

sor3et el ta2deem el 5adamat lel customers .. 3shan kol increment bye7sal beytalla3 output a3la fel wazayef el awlaweya lel 3ameel

y3ni law kan el product calculator bas msh bye7seb sin w cos .. wel customer mohandes .. y3ni el functionality priority (wazayef el awlaweya) belnesbalo lel calculator enno ye7seb sin w cos .. fa bey2ollak hena b2a en kol increment betwassal lel customer awlaweya a3la ..

• **User engagement with the system. Users have to be involved in the development which means the system is more likely to meet their requirements and the users are more committed to the system.**

el user byeb2a mortabet w moshtarek ma3 el system .. y3ni el users beysharko fel develop .. w da ma3 nah en el system bey7a2a2 el mataleb btaree2a a7sn w en el users byeb2o moltazemeen bel system ..

fakes ra3'y hena mel a5er bey2ollak
1- users engagement with system
2- users involve in development
3- users are more committed to the system
3- system meet users requirements more likely

Problems with incremental development

3yobo

1- Management problems

– Progress can be hard to judge and problems hard to find because there is no documentation to demonstrate what has been done.

awel no3 mn el problems .. moshkelet el edara

el taqaddom fel product sa3b enena no7kom 3aleh wel mashakel byeb2a sa3b enena nela2iha 3shan msh bye7sal 3amal documentation 3shan l2esbaat ma tam entago ...

1- progress hard to judge
2- problems hard to find

3shan no documentation 3shan ye3mel demonstrate (yesbet) what has been done

w da tb3n mn el mshakel el edareya l2en el beydeero mashro3 el product lazem yb3rafo eh el tam engazo w eh el mshakel el 7salet ..

2- Contractual problems

– The normal contract may include a specification; without a specification, different forms of contract have to be used.

tani moshkela .. mshakel ta3aqodeya .. y3ni mshakel lma bye7sal en customer yet3a2ed ma3 el sherka el 3amla el program dah

el 3a2d aw el contract el tabee3i momken yb2a m3ah el specification aw mowasafat el program .. mn 3'er el specification de bnedtar nesta5dem contracts kteer b forms kteer w de 7aga msh 7lwa..

3- Validation problems

– Without a specification, what is the system being tested against?

talet moshkela moshkelet el ta72ee2 mn se77et el bernameg
ana law customer raye7 ashteri el bernameg dah .. whwa ma3hoosh specification aw mwasafat leeh .. tb efred ana 3ayz a3mello test aw agarabo .. hagarabo 3la ehh maho asln ana ma3rafsh el bernameg dah ma3mol l2eh wla bye3mel eh .. fa kda b2a 3andi validation problem .. l2eni mba2etsh mbayen lel customer 7aga t5aleeh yet2aked mn se77et el bernameg dah

4- Maintenance problems

– Continual change tends to corrupt software structure making it more expensive to change and evolve to meet new requirements.

rabe3 moshkela heya moshkelet el seyana .. bema enana kol shwaya han3'ayar fel bernameg wn7adeso mn version adeem le version gdeed da olna abl kda enno beycorrubt aw beyefsed el structure bta3 el software w da bey5alli el bernameg dah 3'aly aktar 3shan kol marra bte3mello ta3'yeer w fnafs l wa2t ta5od balak mn el software structure maye7salloosh corruption w law 7asal t3mello maintenance

Agile methods

• Dissatisfaction with the overheads involved in design methods led to the creation of
agile methods. These methods:
– Focus on the code rather than the design;
– Are based on an iterative approach to software development;
– Are intended to deliver working software quickly and evolve this quickly to meet changing requirements.

• Agile methods are probably best suited to small/medium-sized business systems or
PC products.

Fe 7aga esmaha (3.1 Principles of agile methods) page 2 f lecture 6 ..

(exam 2011-2012)

- **Name two XP practices and give a short description of each?**
- **Explain how these practices are related to Agile principles?**

//msh fahem el so2alen dol

Awel 7aga el XP Practices (extreme programming practices) .. e5tar etneen men dol

Incremental planning	**Requirements are recorded on Story Cards and the Stories to be included in a release are determined by the time available and their relative priority. The developers break these Stories into development 'Tasks'.**
Small Releases	**The minimal useful set of functionality that provides business value is developed first. Releases of the system are frequent and incrementally add functionality to the first release.**
Simple Design	**Enough design is carried out to meet the current requirements and no more.**
Test first development	**An automated unit test framework is used to write tests for a new piece of functionality before that functionality itself is implemented.**
Refactoring	**All developers are expected to refactor the code continuously as soon as possible code improvements are found. This keeps the code simple and maintainable.**
Pair Programming	**Developers work in pairs, checking each other's work and providing the support to always do a good job.**
Collective Ownership	**The pairs of developers work on all areas of the system, so that no islands of expertise develop and all the developers own all the code. Anyone can change anything.**

Continuous Integration	**As soon as work on a task is complete it is integrated into the whole system. After any such integration, all the unit tests in the system must pass.**
Sustainable pace	**Large amounts of over-time are not considered acceptable as the net effect is often to reduce code quality and medium term productivity**
On-site Customer	**A representative of the end-user of the system (the Customer) should be available full time for the use of the XP team. In an extreme programming process, the customer is a member of the development team and is responsible for bringing system requirements to the team for implementation.**

Tani 7aga 3elaket el XP bel agile principles

- **XP and agile principles**

- **Incremental development is supported through small, frequent system releases.**
- **Customer involvement means full-time customer engagement with the team.**
- **People not process through pair programming, collective ownership and a process that avoids long working hours.**
- **Change supported through regular system releases.**
- **Maintaining simplicity through constant refactoring of code.**

Lecture 7 : Design Patterns

Design Patterns :-

- Singleton, Façade, Adapter
- Definition, purpose, Class Diagram, Application, Code (singleton, Adapter)

(exam 2011-2012)

What is a design pattern? Mention the three basic categories of design patterns?

- **What Is a Design Pattern?**

□ **Design patterns are recurring solutions to software design problems you find again and again in real-world application development. Patterns are about design and interaction of objects, as well as providing a communication platform concerning elegant, reusable solutions to commonly encountered programming challenges.**

□ **One thing expert designers know *not* to do is solve every problem from first principles. Rather, they reuse solutions that have worked for them in the past. When they find a good solution, they use it again and again.**

□ **Simply, design patterns help a designer get a design "right" faster.**

- **Design Pattern Catalog**

□ **The Gang of Four (GoF) patterns are generally considered the foundation for all other patterns.**

 Design patterns were originally grouped into the categories: creational patterns, structural patterns, and behavioral patterns.

1- Creational Patterns:

are design patterns that deal with object creation mechanisms, trying to create objects in a manner suitable to the situation.

The basic form of object creation could result in design problems or added complexity to the design. Creational design patterns solve this problem by somehow controlling this object creation.

2- Structural Patterns:

Structural patterns are concerned with how classes and objects are composed to form larger structures.

They are design patterns that ease the design by identifying a simple way to realize relationships between entities

3- Behavioral Patterns :

are design patterns that identify common communication patterns between objects and realize these patterns. By doing so, these patterns increase flexibility in carrying out this communication.

(el gay el doctor 2al 3aleh)

(Creational Patterns)

Singleton → A class of which only a single instance can exist

(Structural Patterns)

Adapter → Match interfaces of different classes

Facade → A single class that represents an entire subsystem

(Exam 2011-2012)

What is the difference between facade and decorator design patterns?

Façade → Provide a unified interface to a set of interfaces in a subsystem. Façade defines a higher-level interface that makes the subsystem easier to use.

Decorator → Attach additional responsibilities to an object dynamically. Decorators provide a flexible alternative to subclassing for extending functionality.

Kda yb2a fadel fel lecture de mn 7aget el revision :
Class Diagram & Application & Code bta3 el singleton wel Adapter)